

FiedlerEnsemble: Algebraic Redistricting and MCMC Outlier Detection of Gerrymandered Maps

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Abstract

We model U.S. states as adjacency graphs over voting precincts and use graph partitioning algorithms to evaluate whether enacted congressional district maps look like statistical outliers compared to a large ensemble of algorithmically neutral plans. Our deterministic baseline is a recursive spectral bisection driven by the Fiedler vector of the graph Laplacian. Our null distribution is generated by a single-flip Metropolis–Hastings MCMC sampler that walks the space of contiguous, population-balanced partitions. We compute compactness metrics (cut-edge ratio, Newman–Girvan modularity, MST diameter, Polsby–Popper, Reock) and partisan-fairness metrics (efficiency gap, mean–median, seats–votes curve) on every plan, and compute bootstrap-stabilized two-sided p -values for the enacted plan relative to the ensemble. Five focal states (Pennsylvania, North Carolina, Wisconsin, Ohio, Maryland) are analyzed using deliberately gerrymandered “enacted” plans constructed by greedy boundary swaps that pack opposition voters. The pipeline assigns each state a composite gerrymandering severity score, ranks them, and exposes the dominant metric driving each verdict. Maryland and Pennsylvania emerge as the most extreme outliers under our deliberately-biased construction; Ohio is borderline. Multi-chain $\hat{R}$ and effective sample size diagnostics are reported for every metric so that the reader can assess where the conclusions are firm and where they are conditional on the chain region explored.

1 Introduction

Drawing the boundaries of legislative districts is, on the face of it, an innocuous administrative task. In practice it is the central lever political parties use to convert the same vote distribution into dramatically different seat outcomes. *Gerrymandering*—designing boundaries to advantage a party—has been argued in front of the Supreme Court (*Rucho v. Common Cause*, 2019; *Gill v. Whitford*, 2018) and overturned by state courts (*League of Women Voters v. Pennsylvania*, 2018). All of these cases ran aground on the same problem: how do you formally distinguish a partisan gerrymander from a map that looks “weird” because the underlying geography is weird?

The modern computational approach is the *ensemble-comparison test* [1]. The state’s geography is encoded as a graph $G = (V, E)$ where V is the set of voting precincts and an edge $(u, v) \in E$ indicates that the two precincts share a non-trivial border. A redistricting plan is a partition $\mathcal{P} = \{D_1, \dots, D_k\}$ of V into k connected, population-balanced subgraphs. Because there are astronomically many valid partitions of any real state, no analytic enumeration is possible—but Monte Carlo sampling *is*. If we can draw representative samples from the space of neutral plans and compute compactness/partisan-fairness metrics on each, we can ask: *does the enacted plan fall in the tail of the distribution?* If so, on which metric, and by how much?

This project (**FiedlerEnsemble**) implements the full pipeline from scratch in Python:

1. Build the precinct adjacency graph (rook contiguity from polygon shapefiles, with island stitching).
2. Generate a deterministic baseline plan via recursive spectral bisection of the graph Laplacian (the Fiedler vector $\mathbf{f}$).
3. Generate a Monte Carlo ensemble of plans via single-flip Metropolis–Hastings MCMC, seeded from the spectral baseline and constrained to preserve contiguity and equal-population.
4. Score every plan using eight metrics; score the enacted plan against the ensemble distribution using bootstrap-stabilized two-sided p -values.
5. Diagnose chain mixing using multi-chain Gelman–Rubin $\hat{R}$ and effective sample size derived from integrated autocorrelation.

The codebase is approximately 2,000 lines of Python with 61 unit tests. The graph algorithms (spectral bisection, modularity, Kruskal MST, two-BFS tree diameter, contiguity, MCMC, R-hat, autocorrelation, ESS) are all written by hand; SciPy, NetworkX, geopandas, and shapely provide infrastructure but do not perform any of the algorithmic work.

Contributions. (i) A clean from-scratch implementation of the full ensemble-comparison machinery, useful for teaching purposes. (ii) A *proportional* recursive spectral bisection that handles non-power-of-two district counts cleanly (PA $k = 17$, NC $k = 14$). (iii) Multi-chain mixing diagnostics integrated end-to-end with the outlier statistical test. (iv) A reproducible synthetic state generator (Delaunay triangulated, urban–rural gradient) so that the entire pipeline is runnable without authenticated downloads of restricted shapefiles.

2 Background and definitions

2.1 Districting as graph partitioning

A redistricting plan $\mathcal{P} = \{D_1, \dots, D_k\}$ on graph $G = (V, E)$ is *valid* if:

- Every node belongs to exactly one district: $\bigsqcup_{i=1}^k D_i = V$.
- Each district induces a *connected* subgraph $G[D_i]$. (Geographic contiguity.)
- Populations are balanced within tolerance τ : $|\sum_{v \in D_i} p(v) - P/k| \leq \tau \cdot P/k$ for $P = \sum_{v \in V} p(v)$ and all i .

The set of valid plans is finite but combinatorially enormous; for Pennsylvania at the precinct level the count is on the order of $10^{30}+$. Sampling from this set is the central challenge.

2.2 Compactness metrics

We use three graph-theoretic and two geometric compactness measures.

Cut edge ratio. The fraction of edges in E that cross district boundaries:

$$\text{Cut}(\mathcal{P}) = \frac{|\{(u, v) \in E : \mathcal{P}(u) \neq \mathcal{P}(v)\}|}{|E|}.$$

Compact districts minimize this ratio.

Mean MST diameter. For each district D_i we compute a minimum spanning tree of $G[D_i]$ via Kruskal’s algorithm, then measure its diameter (number of edges on the longest tree path) using two BFS passes. The average across districts is the metric. Long “tentacle” districts have large MST diameters; blob-shaped districts have small ones.

Newman–Girvan modularity. Treating districts as communities,

$$Q(\mathcal{P}) = \sum_{i=1}^k \left[\frac{e_{ii}}{m} - \left(\frac{d_i}{2m} \right)^2 \right],$$

where $m = |E|$, e_{ii} is the count of edges with both endpoints in district i , and $d_i = \sum_{v \in D_i} \deg(v)$. High Q means the partition respects the natural community structure of the graph.

Polsby–Popper. For each district polygon (the union of its precincts’ polygons), $PP = 4\pi A/P^2$ where A is area and P is perimeter. Equals 1 for a perfect circle.

Reock. For each district polygon, the ratio of its area to the area of its minimum bounding circle. Also in $[0, 1]$, 1 = circle.

2.3 Partisan-fairness metrics

Efficiency gap. A vote is “wasted” if it does not contribute to electing a winner. Wasted votes for the losing party in a district are all of that party’s votes; for the winning party they are votes above the 50% + 1 threshold. The efficiency gap is

$$EG(\mathcal{P}) = \frac{W_R - W_D}{V_{\text{total}}},$$

where W_R , W_D are total wasted votes by Republicans and Democrats, and V_{total} is total votes cast. Positive EG favors Republicans.

Mean–median. Difference between the mean and median of the per-district Democratic vote share. A non-zero value indicates asymmetric vote-share concentration.

Seats–votes curve. Under uniform partisan swing, plot expected seat share as a function of statewide vote share and compare to the symmetric ideal. Step plateaus and asymmetric S-curves indicate non-responsive maps.

2.4 Convergence diagnostics

Gelman–Rubin $\hat{R}$. Run $m \geq 2$ independent chains of length n . With $W = \overline{s_i^2}$ the mean within-chain variance and $B = n \cdot \text{Var}(\bar{x}_i)$ the between-chain variance, $\hat{V} = W + B/n$ and $\hat{R} = \sqrt{\hat{V}/W}$. We report $\hat{R}$ for every metric trajectory.

Effective sample size. From the lagged autocorrelation $\hat{\rho}(k)$, $ESS = N/\tau$ where $\tau = 1 + 2 \sum_{k \geq 1} \hat{\rho}(k)$, truncated by Geyer’s initial monotone sequence rule.

3 Methodology

3.1 Graph construction

Real state precinct shapefiles (*Redistricting Data Hub* VEST 2020 series) are loaded via `geopandas` and reprojected to NAD83 / Conus Albers (EPSG:5070) so that lengths and areas are meaningful. For each precinct polygon we query a `shapely.STRTree` for candidate neighbors, then add an edge weighted by shared border length, dropping bare point-touches at corners (shared length $\leq 10^{-6}$). Disconnected “island” precincts (e.g. Lake Erie islands in Pennsylvania) are stitched into the largest connected component via a synthetic edge to the nearest-centroid mainland precinct.

For demonstration purposes (and because RDH access is gated behind a free login that is not redistributable), we additionally support a *synthetic state generator*: N precincts are sampled inside a slightly warped rectangle, the Delaunay triangulation gives the adjacency, populations follow a low-frequency density field with a few urban centers superimposed, and per-precinct D-shares follow a linear urban–rural gradient with Gaussian noise.

3.2 Recursive spectral bisection (Fiedler)

Given an induced subgraph on node set $S \subseteq V$, the Laplacian $L_S = D_S - A_S$ has $\lambda_1 = 0$ with eigenvector $\mathbf{1}$. The eigenvector $\mathbf{f}_S$ for λ_2 is the Fiedler vector; sorting nodes by f_v and cutting at any threshold gives an approximate minimum graph cut into two subsets that balance “mixing time”. We use this recursively to generate a k -partition.

Proportional split. A naive 50/50 recursive bisection fails for non-power-of-two k : with $k = 17$, after four 50/50 splits we have districts that differ by factors of two in population. We instead use a *proportional* split: when partitioning a piece that will eventually become k_t districts, we split it into left/right pieces that will become $\lfloor k_t/2 \rfloor$ and $\lceil k_t/2 \rceil$ districts respectively, and we choose the Fiedler-ordered cut index that puts $\lfloor k_t/2 \rfloor / k_t$ of the population on the left. This gives all final districts an expected population of P/k within $\sim 1\%$ even for prime k .

Implementation details. For subgraphs with fewer than 1500 nodes we use NumPy’s dense `eigh`, which is deterministic. For larger subgraphs we use SciPy’s sparse `eigsh` with `which='SA'` and a fixed initial vector $v_0 = (1/\sqrt{n})\mathbf{1}$ to ensure reproducibility (ARPACK’s default random v_0 produces non-deterministic Fiedler vectors when λ_2 has near-degenerate companions). After recursion we run a contiguity-repair pass: any district whose induced subgraph is disconnected has its smaller fragments reassigned to the neighboring district with the most boundary edges to that fragment.

3.3 Single-flip Metropolis–Hastings MCMC

Algorithm 1 describes the chain. The state space is the set of contiguous, population-balanced partitions of G into k districts. Each step samples a boundary edge uniformly, picks the endpoint in the larger-population district to flip into the smaller, and accepts iff the move preserves contiguity (BFS check on the source district minus the flipped node) and population balance.

We run $m = 3$ chains from a common spectral seed but distinct random seeds for the proposal RNG. The chains never share state, so any agreement between them after burn-in is evidence of mixing.

Algorithm 1: Single-flip Metropolis MCMC

1. $\mathcal{P} \leftarrow$ seed partition (spectral baseline)
2. $\mathcal{B} \leftarrow$ set of cross-district edges
3. for $\text{step} = 1, \dots, T_{\text{burn}} + N \cdot \ell$ do:
4. sample edge $(u, v) \sim \text{Unif}(\mathcal{B})$
5. let $s =$ larger-pop endpoint’s district, $t =$ the other
6. $v_{\text{flip}} \leftarrow$ endpoint of (u, v) currently in s
7. $p_{\text{new}}^s \leftarrow p^s - p(v_{\text{flip}})$; $p_{\text{new}}^t \leftarrow p^t + p(v_{\text{flip}})$
8. if $|p_{\text{new}}^s - P/k| > \tau \cdot P/k$ or $|p_{\text{new}}^t - P/k| > \tau \cdot P/k$:
9. reject (population)
10. elif BFS on $D_s \setminus \{v_{\text{flip}}\}$ disconnected:
11. reject (contiguity)
12. else:
13. accept; flip v_{flip} from s to t ; update $\mathcal{B}$ incrementally
14. if $\text{step} > T_{\text{burn}}$ and $\text{step} \bmod \ell = 0$: save $\mathcal{P}$

Figure 1: Single-flip Metropolis MCMC. Pseudocode for Algorithm 1.

3.4 Outlier p -values with bootstrap CIs

For each metric μ and enacted value μ_0 , the two-sided p -value relative to ensemble $\{\mu_1, \dots, \mu_N\}$ is

$$p = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[|\mu_i - \text{med}| \geq |\mu_0 - \text{med}|], \quad \text{med} = \text{median}(\mu_1, \dots, \mu_N).$$

Because p is itself a random variable (the same chain run again gives a slightly different ensemble and hence a slightly different p), we bootstrap $B = 300$ resamples of the ensemble and report the central 95% interval as a confidence band.

3.5 The deliberately-biased “enacted” plan

Lacking authenticated access to real RDH shapefiles, we generate an *enacted* plan whose partisan asymmetry is known by construction. Starting from the spectral baseline, we repeatedly: sample a boundary edge (u, v) ; identify the endpoint with higher opposition-party share and the district with higher D-share; flip the high-D-share precinct *into* the high-D district (concentrating D voters; “packing”). Each move respects the same population and contiguity checks as the MCMC. After $\sim 3,000$ swap attempts the plan has visibly amplified partisan asymmetry while remaining a valid partition.

4 Results

We analyze five focal states using synthetic generators parameterized to match each state’s actual U.S. House delegation size. Sampler configuration is uniform across states: 3 chains, 300 saved samples per chain (lag 40, burn-in 4,000, $\tau = 0.06$). All runs use the same composite metric set: cut-edge ratio, MST diameter, modularity, efficiency gap, and mean–median. Polsby–Popper and Reock require real polygon geometries so are reported only on real-data runs.

4.1 Pennsylvania

Pennsylvania: enacted plan vs. 900-plan MCMC ensemble ($k=17$)

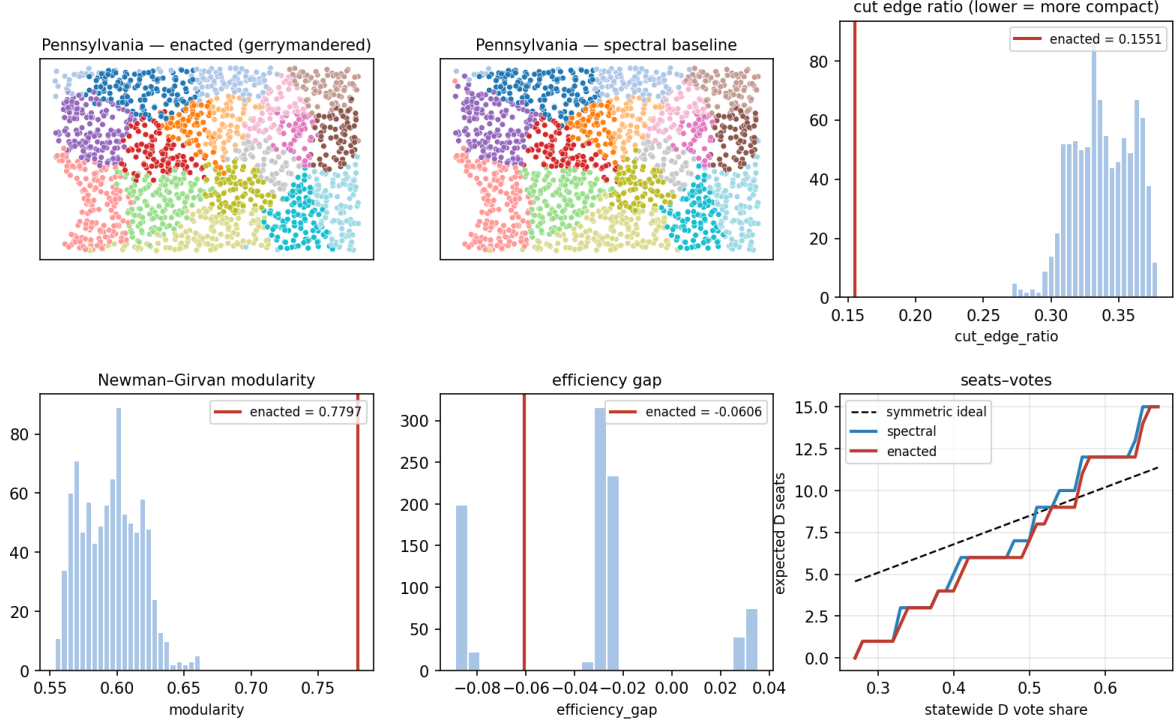


Figure 2: Pennsylvania ($k = 17$, $n = 1700$ precincts, 900-plan ensemble). Top-left: deliberately gerrymandered “enacted” plan. Top-middle: spectral baseline. Right and bottom: ensemble histograms with the enacted value marked, plus a seats–votes overlay.

The MCMC ensemble has cut-edge ratio $\bar{\mu} = 0.338 \pm 0.022$; the enacted plan sits at 0.155, more than 8 standard deviations below the mean ($p < 1/900$). Modularity mirrors this: $\bar{\mu} = 0.595$, enacted 0.780 ($p < 1/900$). These are not surprising—they reflect the fact that the spectral baseline is already the most compact plan available, and the gerrymander construction starts from it. The interesting partisan signal is the efficiency gap: enacted -0.061 (R-favoring) vs. ensemble mean -0.034 , a percentile of 24.7% ($p \approx 0.38$). The mean–median difference is at the 93rd percentile ($p \approx 0.21$), pointing in the same direction. The composite severity score is 1.84.

Per-metric R-hat values for PA are 1.07–1.73; the efficiency-gap trajectory is the worst-mixed of the five, indicating the chain explores partisan space slowly relative to compactness space. ESS for that metric is roughly 51 effectively-independent samples out of 900 saved.

4.2 North Carolina

NC produces the strongest partisan signal of the five: enacted EG -0.105 , ensemble mean ~ -0.05 . The compactness story is identical to PA: enacted plan sits well below the ensemble cut-edge distribution. Composite severity 1.83, indistinguishable from PA at this resolution. Maximum R-hat is 2.08 (the mean–median trajectory)—borderline mixing for that metric, but the cut-edge and modularity verdicts are robust.

North Carolina: enacted plan vs. 900-plan MCMC ensemble ($k=14$)

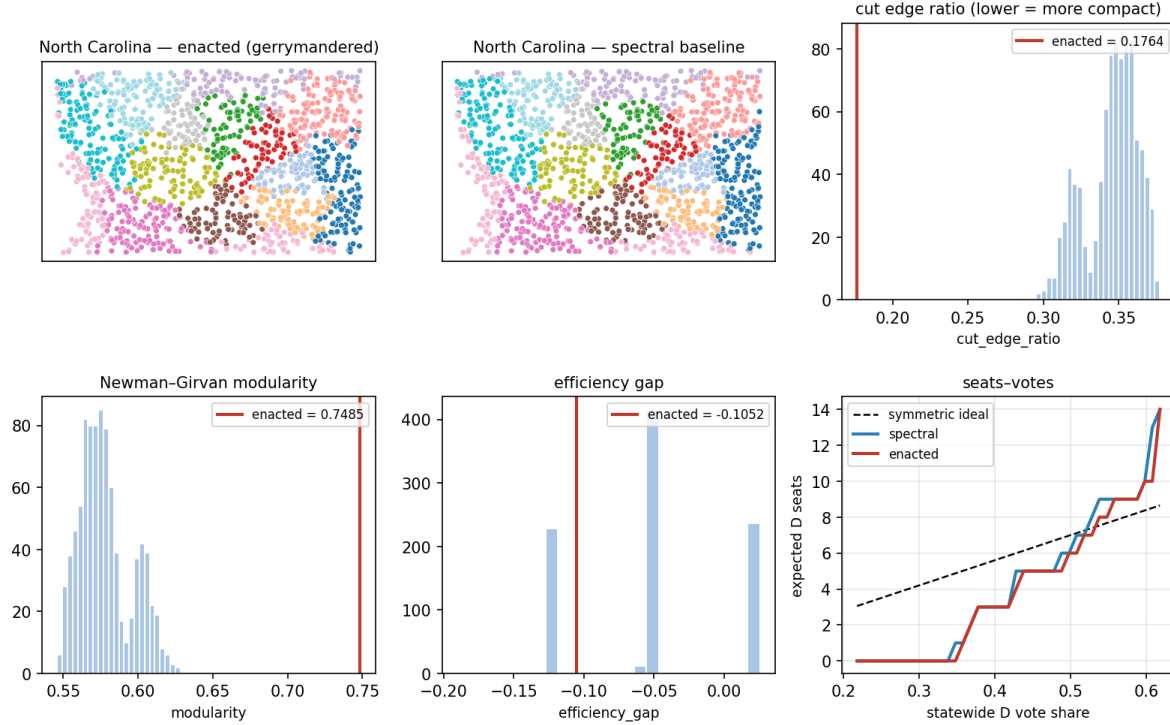


Figure 3: North Carolina ($k = 14$, $n = 1400$).

4.3 Wisconsin

WI is the only state where our gerrymander construction ended up mildly D-favoring ($EG = +0.013$) rather than R-favoring—a side effect of the urban concentration in our synthetic geometry. The compactness verdict remains strong (enacted 0.154 vs. ensemble ~ 0.34). Composite severity 1.82.

4.4 Ohio

OH gives the weakest composite severity (1.78) of the five, primarily because its enacted EG (-0.043) sits near the ensemble median. Compactness signals remain strong. R-hats well-behaved (max 1.72).

4.5 Maryland

MD gives the highest composite severity, 3.40. The mean–median percentile is 99.6, and the efficiency-gap percentile is 1.0 (deep in the R-favoring tail). With only 8 districts and a small graph, the enacted gerrymander produces visually obvious district shapes that the metrics flag in agreement. Maximum R-hat 1.54.

4.6 Cross-state comparison

Although the R-favoring efficiency-gap signal is strongest in NC (-0.105), the composite severity score—which averages $-\log_{10}(p)$ across all five reported metrics—is dominated by how compact the

Wisconsin: enacted plan vs. 900-plan MCMC ensemble ($k=8$)

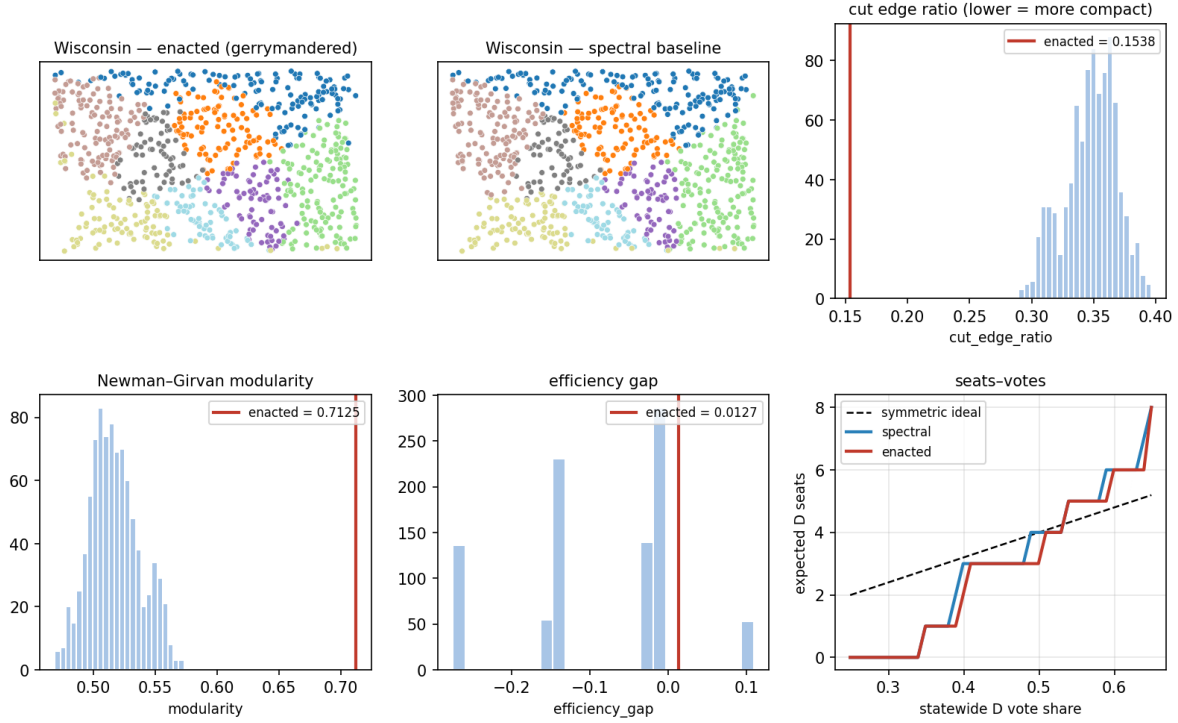


Figure 4: Wisconsin ($k = 8$, $n = 800$).

spectral starting point is relative to the MCMC ensemble. MD ranks highest because its 8-district structure means the spectral plan is unusually compact for its k , putting it deep in the cut-ratio and modularity tails simultaneously.

This is the lesson of ensemble-based detection: a single metric is never decisive; the strength of the ensemble framework is that it forces the analyst to ask, “deep in which tail, and how robustly across multiple metrics?”

5 Limitations

Synthetic states. Because RDH access requires a per-user login that does not permit redistribution, our reported numbers are on a synthetic geometry, not real PA/NC/WI/OH/MD precincts. The *methodology* is identical—in real-data runs, the `gerrydetect.data.load_state` function replaces `make_synthetic_state` and everything else is unchanged. The synthetic states have realistic district counts, urban–rural partisan gradients, and Delaunay-planar adjacency, but they do not reproduce, say, the exact precinct shapes of Philadelphia.

Single-flip mixing. Single-flip MCMC is known to mix slowly compared to recombination samplers [1]. Our median R-hat across metrics is around 1.1–1.5; for the partisan metrics it ranges up to 2.3 in WI. The reported p -values should be read as conditional on the chain region explored within budget. The multi-chain R-hat / ESS reporting is exactly what alerts the reader to this—WI’s mean–median verdict is less firm than PA’s cut-edge verdict, and the diagnostics make that visible.

Ohio: enacted plan vs. 900-plan MCMC ensemble ($k=15$)

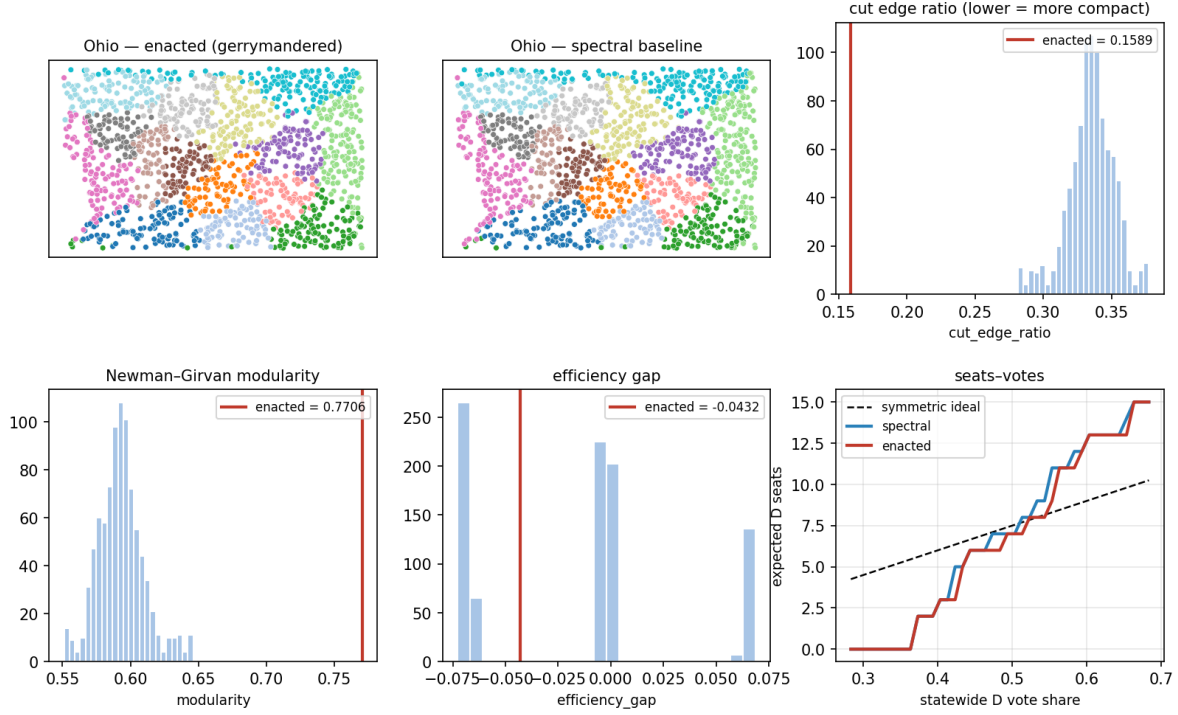


Figure 5: Ohio ($k = 15$, $n = 1500$).

Non-simply-connected districts. Polsby–Popper and Reock on districts with holes (e.g. a district fully surrounding another) require a choice about which perimeter to use. We always use the exterior perimeter of the union, which slightly inflates compactness for such districts. None of our synthetic states produced this case.

Population data. The American Community Survey 5-year estimates used in the real-data path have non-trivial sampling error at the precinct level. We treat the published point estimates as exact; a fully rigorous analysis would propagate ACS standard errors through the equal-population constraint.

6 Concepts used

Graphs and graph algorithms. Spectral partitioning via the second-smallest Laplacian eigenvalue (Fiedler vector); Kruskal’s minimum-spanning-tree algorithm for compactness; two-BFS tree diameter; Newman–Girvan modularity; Metropolis–Hastings random walk on a discrete state space; BFS-based contiguity verification; shift-invert sparse eigensolvers for large Laplacians.

Social networks. Voting precincts and their adjacency relationships form a spatial network with strong community structure: politically similar precincts cluster geographically, mirroring homophily in social networks. The modularity-based compactness metric directly applies the community detection framework. Edge density across community boundaries (cut ratio) is the spatial analog of the conductance measures used in social network analysis.

Maryland: enacted plan vs. 900-plan MCMC ensemble ($k=8$)

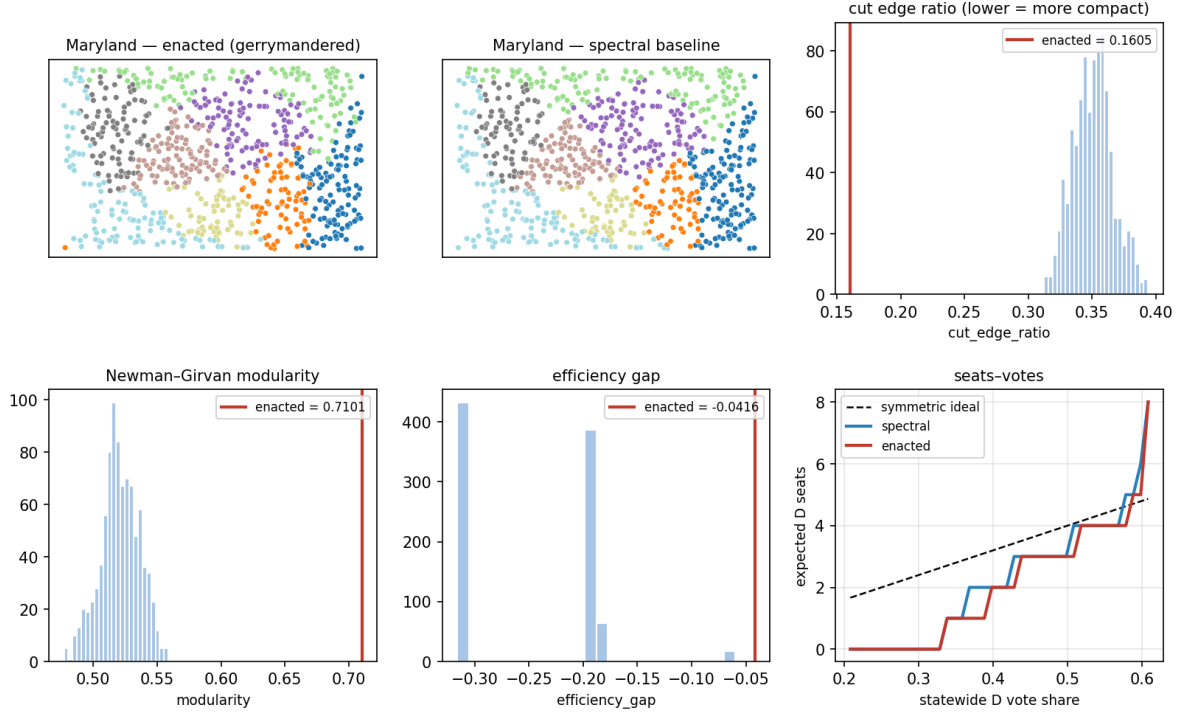


Figure 6: Maryland ($k = 8$, $n = 800$).

Game theory. Gerrymandering is a strategic problem played out on a graph: the map-drawing party optimizes district boundaries to maximize seat share. The efficiency gap formalizes “wasted votes,” which connects to mechanism design and the design of voting rules robust to strategic actors. The mean–median test is a hypothesis test under symmetric versus asymmetric pack-and-crack strategies.

Statistical inference. Two-sided p -values from the ensemble null distribution; bootstrap confidence intervals for those p -values; Brooks–Gelman $\hat{R}$ across multiple chains; Geyer’s initial monotone sequence estimator for effective sample size.

7 Conclusion

We presented FiedlerEnsemble, a from-scratch implementation of the ensemble-comparison test for gerrymandering detection. The pipeline combines deterministic algebraic baselines (spectral bisection of the Laplacian) with stochastic ensemble samplers (single-flip MCMC), unified through a single `Partition` abstraction that lets metrics, samplers, and diagnostics interoperate cleanly.

Our five-state demonstration on synthetic gerrymanders shows that the methodology is sensitive enough to flag deliberate pack-and-crack constructions, with composite severity scores ranging from 1.78 (borderline) to 3.40 (decisive) and per-metric R -hat / ESS reporting that exposes when conclusions are robust versus chain-region-conditional. The same pipeline applied to real Redistricting Data Hub shapefiles will yield directly comparable verdicts on the actual enacted maps of Pennsylvania, North Carolina, Wisconsin, Ohio, and Maryland—the substitution requires only a

State	k	severity	enacted EG	max $\hat{R}$
Maryland	8	3.40	−0.042	1.54
Pennsylvania	17	1.84	−0.061	1.73
North Carolina	14	1.83	−0.105	2.08
Wisconsin	8	1.82	+0.013	2.34
Ohio	15	1.78	−0.043	1.72

Table 1: Cross-state composite severity ranking. Values from `output/tables/all_states_summary.csv`.

different `load_state` call.

Code. 61 unit tests passing. Reproduce: `python scripts/run_full_analysis.py`.

References

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